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Erdős-Ko-Rado theorems on the weak Bruhat lattice

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Abstract

Let $\mathscr{L} = (X, \preceq)$ be a lattice. For $\mathcal{P} \subseteq X$ we say that $\mathcal{P}$ is *t-intersecting* if $\mathsf{rank}(x \wedge y) \geq t$ for all $x, y \in \mathcal{P}$. The seminal theorem of Erdős, Ko and Rado describes the maximum intersecting $\mathcal{P}$ in the lattice of subsets of a finite set with the additional condition that $\mathcal{P}$ is contained within a level of the lattice. The Erdős-Ko-Rado theorem has been extensively studied and generalized to other objects and lattices.

In this paper, we focus on intersecting families of permutations as defined with respect to the weak Bruhat lattice. In this setting, we prove analogs of certain extremal results on intersecting set systems. In particular we give a characterization of the maximum intersecting families of permutations in the Bruhat lattice. We also characterize the maximum intersecting families of permutations within the rth level of the Bruhat lattice of permutations of size n, provided that n is large relative to r.

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1 Introduction

Let $\mathscr{L} = (X, \preceq)$ be a lattice. For $\mathcal{P} \subseteq X$ we say that $\mathcal{P}$ is *t-intersecting* if $\mathsf{rank}(x \wedge y) \geq t$ for all $x, y \in \mathcal{P}$. If $\mathscr{L}$ is the subset lattice, two subsets A, B are t-intersecting exactly when $|A \cap B| \geq t$ (we will refer to 1-intersecting sets simply as intersecting sets). The problem of finding maximum collections of intersecting sets has a long history, see [3] and the references within. Perhaps the most famous result is the Erdős-Ko-Rado (EKR) Theorem [2]. This theorem gives the size of the largest collection of sets at level r in the subset lattice such that any two of the sets intersect.

Theorem 1.1 *Let r and n be integers with $n \geq 2r$. If $\mathcal{A}$ is an intersecting collection of r-subsets of the set $\{1, \ldots, n\}$ then*

$$|\mathcal{A}| \leq \binom{n-1}{r-1}.$$

Moreover, if $n > 2r$ equality holds if and only if $\mathcal{A}$ consists of all the r-subsets that contain a common point.

Another well-known result is Katona's theorem [7], which gives the size of the largest collection of t-intersecting sets (with no restriction on the size of the sets) in the subset lattice on n points. If $n + t = 2v$, then this maximum size is achieved by the collection of all subsets with size larger than v. If $n + t = 2v - 1$, an example of a collection of maximum size is the collection that contains all subsets with size at least v, along with all subsets of size $v-1$ that do not contain a fixed element. In the special case that $t = 1$ these sets have size 2^{n-1}. This implies that the collection of all subsets that contain a fixed element is a 1-intersecting set of maximum size. In this case there are also many other (non-isomorphic) collections of maximum size.

One of the reasons that the EKR Theorem is so famous is that versions of it hold for many other objects; for example there are versions for integer sequences, vector spaces over a finite field, perfect matchings, independent sets in graphs, and permutations (these are just a few of the examples, again see [3], and the references within, for more details).

In this paper, we give versions of Katona's Theorem and the Erdős-Ko-Rado Theorem for the weak Bruhat lattice of permutations. We start with some general results for all lattices. Next we define and give properties of the weak Bruhat lattice. Section 4 has a characterization for the maximum intersecting collections of intersecting permutations in the Bruhat lattice. Section 5 gives a version of the EKR theorem for permutations in level r of the Bruhat lattice, provided that n is large relative to r. In the sequel, collections of sets will be called *systems*, and collections of permutations will be called *families*.

2 General lattices

For a subset $\mathcal{F}$ of elements in a lattice $\mathscr{L} = (X, \preceq)$, define the *upset of* $\mathcal{F}$ to be

$$\mathsf{up}(\mathcal{F}) = \{z \in X \mid x \preceq z, \text{ some } x \in \mathcal{F}\}.$$

Similarly, the *downset of* $\mathcal{F}$ is defined to be

$$\mathsf{down}(\mathcal{F}) = \{z \in X \mid z \preceq x, \text{ some } x \in \mathcal{F}\}.$$

For any $\mathcal{F}$, if $\mathsf{up}(\mathcal{F}) = \mathcal{F}$, then we say that $\mathcal{F}$ is an *upset*. Similarly, $\mathcal{F}$ is a *downset* if $\mathsf{down}(\mathcal{F}) = \mathcal{F}$. Upsets and downsets are also called filters and ideals, respectively.

For a single element $p \in \mathscr{L}$, the family $\mathsf{up}(\{p\})$ (which we abbreviate $\mathsf{up}(p)$) is called a *star* with *center* p. If the rank of p is t, then $\mathsf{up}(p)$ is t-intersecting; we call $\mathsf{up}(p)$ a *canonical* t-intersecting family. More generally, for some $\mathcal{H} \subseteq \mathscr{L}$ we define $\mathsf{up}_{\mathcal{H}}(\mathcal{F}) = \mathsf{up}(\mathcal{F}) \cap \mathcal{H}$. In this paper, we only consider the case where $\mathcal{H}$ is the set of all elements in $\mathscr{L}$ with rank k; this set is denoted by $\mathscr{L}_k$. If $\mathsf{rank}(p) = t$, then $\mathsf{up}_{\mathscr{L}_k}(p)$ is called a *t-star at level k*.

We say that $\mathscr{L}$ has the *t-EKR property* (or is *t-EKR*) if a t-star is a largest t-intersecting family in $\mathcal{L}$. Define $f_t(\mathscr{L})$ to be the maximum size of a t-intersecting family from the lattice $\mathscr{L}$, then $\mathscr{L}$ has the t-EKR property if $f_t(\mathscr{L}) = \max_{p \in \mathscr{L}_t} |\mathsf{up}(p)|$. A lattice that has 1-EKR property will simply be said to have the EKR property.

Similarly, $f_t(\mathscr{L}_k)$ is defined to be the maximum size of a t-intersecting family in level k of the lattice. The level $\mathscr{L}_k$ has the *t-EKR property* (or *is t-EKR*) if the size of the largest t-intersecting family of $\mathscr{L}_k$ can be realized by a t-star at level k. This is equivalent to saying that $f_t(\mathscr{L}_k) = \max_{p \in \mathscr{L}_t} |\mathsf{up}_{\mathcal{L}_k}(p)|$. Again, we suppress the use of t when $t = 1$.

We say that a lattice $\mathscr{L}$ is *uniquely complemented* if, for all $x \in X$, there exists a unique $y \in X$ such that $\mathsf{rank}(x \wedge y) = 0$ and $\mathsf{rank}(x \vee y) = \mathsf{rank}(\mathscr{L})$. This is a set of lattices which clearly has the EKR property.

Theorem 2.1 *Any uniquely complemented lattice has the EKR property.*

Proof. Assume that $\mathscr{L}$ is a uniquely complemented lattice. Denoting the complement of an element $F \in \mathscr{L}$ by $\overline{F}$, at most one of the pair $\{F, \overline{F}\}$ can belong to an intersecting family. Thus an intersecting set can be no larger than $|\mathscr{L}|/2$.

Let p be any element of the lattice with rank 1, then $|\mathsf{up}(p)| = |\mathscr{L}|/2$. □

Since the subset lattice is uniquely complemented, this theorem implies that the subset lattice has the EKR property. Katona's theorem shows that the subset lattice does not have the t-EKR property for $t > 1$. The Erdős-Ko-Rado Theorem says that levels below $n/2$ have the EKR property. In fact, Ahlswede and Khachatrian's complete

intersection theorem [1] determines for which values of n, k and t, level k has the t-EKR property in the subset lattice for $N = \{1, \ldots, n\}$.

Many of the standard proofs for the EKR property use regularity conditions on the set. Suda [8] considered the t-EKR property for levels in semi-lattices with regularity conditions. The Bruhat lattice is different since does not satisfy these regularity conditions. Specifically the upsets of different elements at the same level may have different sizes, this makes the Bruhat lattice a particularly interesting lattice to consider. More details are given in the next section.

3 Properties of the weak Bruhat lattice

The (right) weak Bruhat ordering on the symmetric group produces a lattice $\mathscr{B}_n = (\mathrm{Sym}(n), \preceq)$. For any permutation $p \in \mathrm{Sym}(n)$ we write $p = p_1 p_2 \cdots p_n$ to mean $p(i) = p_i$ (this is the second line in the two line notation). The covering relation $\prec$ that generates the relation $\preceq$ is defined by $p_1 p_2 \cdots p_n \prec q_1 q_2 \cdots q_n$ whenever there is some $i \in \{1, \ldots, n-1\}$, such that $p_i < p_{i+1}$, $q_i = p_{i+1}$, $q_{i+1} = p_i$, and $q_k = p_k$ otherwise. This means that $p \prec q$ if q is obtained by reversing two consecutive and increasing elements from p. See Figure 1 for $\mathscr{B}_4$. We call the transpositions $(i, i+1)$ for $i \in \{1, \ldots, n-1\}$ the *generators* and will denote the generator $(i, i+1)$ by g_i. We also define $\mathbf{G}_n = \{g_i \mid 1 \leq i < n\}$ to be the *generating set* for $\mathscr{B}_n$. From another point of view, $\mathscr{B}_n$ is the Cayley graph generated by $\mathbf{G}_n$, turned into a lattice with ranks $\mathscr{B}_{n,l}$ given by distance from the identity $\mathbf{id}$.

For a permutation $p \in \mathrm{Sym}(n)$ we define its set of *inverse descents* by

$$\mathsf{ID}(p) = \{g_i \mid i+1 \text{ precedes } i \text{ in } p\},$$

and for a set of permutations $\mathcal{P}$, we define $\mathsf{ID}(\mathcal{P}) = \{\mathsf{ID}(p) \mid p \in \mathcal{P}\}$. The set of inverse descents play an important role, since for any $p \in \mathrm{Sym}(n)$, we have $g_i \in \mathsf{ID}(p)$ if and only if $g_i \preceq p$ in the weak order.

We denote the *inversion set* for a permutation $p \in \mathrm{Sym}(n)$ by

$$\mathsf{Inv}(p) = \{(p_j, p_i) \mid 1 \leq i < j \leq n,\, p_j < p_i\}$$

with this definition, $\mathsf{rank}(p) = |\mathsf{Inv}(p)|$. Our definition of inverse descents is nonstandard; usually the inverse descents are integers, but we are using the simple transpositions indexed by those integers.

The inversion set is unique for each permutation, while the set of inverse descents is not. For example,

$$\begin{aligned} \mathsf{ID}(3214) &= \{g_1, g_2\}, & \mathsf{Inv}(3214) &= \{(1,2),(1,3),(2,3)\}, \\ \mathsf{ID}(3241) &= \{g_1, g_2\}, & \mathsf{Inv}(3241) &= \{(1,2),(1,3),(1,4),(2,3)\}. \end{aligned} \tag{1}$$

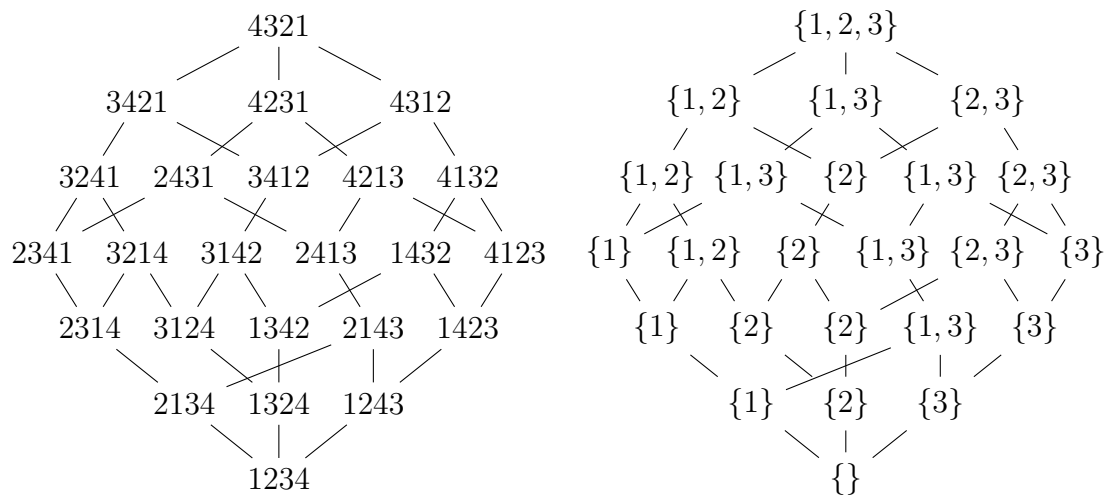

Figure 1: Two representations of the Bruhat lattice $\mathcal{B}_4$. On the left, vertices are labeled by $p = p_1p_2p_3p_4$, where $p(i) = p_i$. On the right, each vertex p is labeled by its set of inverse descents $\mathsf{ID}(p)$.

For a set $A \subseteq \mathbf{G}_n$, define the *multiplicity of* A by $|\{p \in \mathscr{B}_n \mid \mathsf{ID}(p) = A\}|$.

The Bruhat ordering on permutations is an example of a uniquely complemented lattice. The complement of the permutation $p = p_1p_2 \ldots p_n$ is the permutation $\overline{p} = p_n \ldots p_2p_1$. The unique complement has the property that

$$\mathsf{Inv}(\overline{p}) = \{(i,j) \mid 1 \leq i < j \leq n\} \setminus \mathsf{Inv}(p),$$

and also that $\mathsf{ID}(\overline{p}) = \mathbf{G}_n \setminus \mathsf{ID}(p)$.

Since the Bruhat lattice is a uniquely complemented lattice, we have the following corollary of Theorem 2.1. (We alter notation slightly to write $f_t(\mathscr{B}_n)$ as $f_t(n)$ and $f_t((\mathscr{B}_n)_k)$ as $f_{t,k}(n)$.)

Corollary 3.1 *Let n be an integer with $n \geq 2$, then $f_1(n) = n!/2$.*

The goal of the next section is give properties of intersecting families in the Bruhat lattice in order to characterize the extremal intersecting sets.

4 Properties of intersecting Sets in $\mathscr{B}_n$

In this section we give properties of intersecting sets in the Bruhat lattice. The first claim follows directly from the definitions in the previous section.

Claim 4.1 *Let $p, q \in \mathscr{B}_n$, then the following hold.*

1. *If $p \preceq q$, then $\mathsf{ID}(p) \subseteq \mathsf{ID}(q)$.*
2. $\mathsf{ID}(p \wedge q) = \mathsf{ID}(p) \cap \mathsf{ID}(q)$.
3. $p \wedge q \neq \mathbf{id}$ *if and only if* $\mathsf{ID}(p) \cap \mathsf{ID}(q) \neq \emptyset$.
4. $p \preceq q$ *if and only if* $\mathsf{Inv}(p) \subseteq \mathsf{Inv}(q)$.
5. $g_i \in \mathsf{ID}(p)$ *if and only if* $(i, i+1) \in \mathsf{Inv}(p)$.
6. $\mathsf{Inv}(p)$ *is transitively closed; that is, if* $(a, b) \in \mathsf{Inv}(p)$ *and* $(b, c) \in \mathsf{Inv}(p)$, *then* $(a, c) \in \mathsf{Inv}(p)$.

Proof. Statements (1) and (4) follow directly from the definition of the relation in the Bruhat lattice.

Statement 1 implies that $\mathsf{ID}(p \wedge q) \subseteq \mathsf{ID}(p) \cap \mathsf{ID}(q)$. Conversely, if $g_i \in \mathsf{ID}(p) \cap \mathsf{ID}(q)$, then $(i, i+1) \preceq p \wedge q$, and $g_i \in \mathsf{ID}(p \wedge q)$. Thus $\mathsf{ID}(p \wedge q) = \mathsf{ID}(p) \cap \mathsf{ID}(q)$ and the results (2) and (3) follow.

Statements (5) and (6) follow from the definition of the inversion set. □

From the permutations in Example 1, we can see that distinct permutations may have the same set of inverse descents. This implies that converse of Statement (1) in the above claim is not true, since it is possible to have incomparable permutations p and q with $\mathsf{ID}(p) = \mathsf{ID}(q)$. For example, 2431 and 4213 are incomparable, but have the same set of inverse descents.

Lemma 4.2 *For any subset $A \subseteq \mathbf{G}_n$ there exists a unique minimum (in the Bruhat ordering) permutation p such that $\mathsf{ID}(p) = A$.*

Proof. Set $p = \bigvee_{g_i \in A} g_i$, since $\mathscr{B}_n$ is a lattice, p is well-defined and unique. Clearly $\mathsf{ID}(p) = A$. If q is any other permutation with $\mathsf{ID}(q) = A$, then $p \leq q$. □

For example, if $A = \{2, 3, 4, 6\}$, then $p = g_2 \vee g_3 \vee g_4 \vee g_6$. Then the first seven positions of the permutation would be 1432576.

Lemma 4.2 shows that the multiplicity for any inverse descents set is at least 1. This is not true for the inversion set; for example there is no permutation with the inversion set $\{(1, 2), (2, 3)\}$ (if 2 is before 1 and 3 is before 2, then 3 is also before 1). Part (6) of Claim 4.1 implies that $\mathsf{Inv}(\overline{p})$ is transitively closed so we have the following corollary that shows that many inverse descent sets are not possible.

Corollary 4.3 *If $(a,b) \notin \mathsf{Inv}(p)$ and $(b,c) \notin \mathsf{Inv}(p)$, then $(a,c) \notin \mathsf{Inv}(p)$.*

For a subset $\mathcal{F}$ of elements in a lattice $\mathscr{L}$, define

$$\mathsf{min}(\mathcal{F}) = \{p \in \mathcal{F} \mid \text{ if there exists } q \in \mathcal{F} \text{ with } q \preceq p, \text{ then } q = p\ \}.$$

Observe that $\mathcal{F}$ is an antichain if and only if $\mathsf{min}(\mathcal{F}) = \mathcal{F}$. Further, if $\mathcal{F}$ is a maximum t-intersecting set, then $\mathsf{up}(\mathsf{min}(\mathcal{F})) = \mathcal{F}$.

For a subset $A \subseteq \mathbf{G}_n$, let

$$\pi(A) = \mathsf{min}\{p \in \mathscr{B}_n \mid \mathsf{ID}(p) = A\}.$$

Lemma 4.2 shows that this is a unique well-defined permutation; this permutation is called the *minimal* element for A. Claim 4.1 implies the following lemma.

Lemma 4.4 *Let $p = \pi(A)$. Then for any q with $\mathsf{ID}(q) = A$, we have that $\mathsf{Inv}(p) \subseteq \mathsf{Inv}(q)$ and thus that $p \preceq q$.* □

Lemma 4.5 *Let $\mathcal{P}$ be a family of permutations in the Bruhat lattice. Then, the following statements are equivalent.*

1. *$\mathcal{P}$ is intersecting.*
2. *$\mathsf{up}(\mathcal{P})$ is intersecting.*
3. *$\mathsf{min}(\mathcal{P})$ is intersecting.*
4. *$\mathsf{ID}(\mathcal{P})$ is an intersecting set system.*
5. *If $\mathcal{G}$ is a system of subsets of $\mathbf{G}_n$, with $\mathcal{P} = \{\pi(g) \mid g \in \mathcal{G}\}$, then $\mathcal{G}$ is intersecting.*

Proof. Since $\mathscr{B}_n$ is a poset, for all $p, p', q, q' \in \mathcal{P}$ with $p \preceq p'$ and $q \preceq q'$, it follows that $p \wedge q \preceq p' \wedge q'$. This implies that if a set $\mathcal{P}$ is intersecting, then $\mathsf{up}(\mathcal{P})$ is also intersecting. This fact show that Statement (1) implies Statement (2). Statement (2) implies Statement (3) since $\mathsf{min}(\mathcal{P}) \subseteq \mathsf{up}(\mathcal{P})$. Statement (3) implies Statement (1) since $\mathcal{P} \subseteq \mathsf{up}(\mathsf{min}(\mathcal{P}))$. Thus the first three statements are equivalent.

Finally, Statement 3 of Claim 4.1 implies that Statements (4) and (5) are equivalent to (1). □

5 Characterization of extremal intersecting sets

In this section we give a characterization of the maximum intersecting families in $\mathscr{B}_n$. If $\mathcal{P} \subseteq \mathscr{B}_n$ is an intersecting family of permutations of maximum size, then $\mathcal{P} = n!/2$, so for every permutation p, either $p \in \mathcal{P}$ or $\overline{p} \in \mathcal{P}$. An intersecting family $\mathcal{P}$ is called *maximal* if every permutation that intersects all the permutations in $\mathcal{P}$ is also in $\mathcal{P}$; equivalently, there are no elements not in $\mathcal{P}$, that intersect all the elements in $\mathcal{P}$.

For any set system $\mathcal{A}$ defined on an $(n-1)$-set, which we may take to be $\mathbf{G}_n$, there is a corresponding subset of $\mathscr{B}_n$ defined by

$$\mathcal{P}(\mathcal{A}) = \mathsf{up}\left(\bigcup_{A \in \mathcal{A}} \pi(A)\right).$$

This means that $\mathcal{P}(\mathcal{A})$ is the set of all permutations p such that there exists an $A \in \mathcal{A}$ such that $A \subseteq \mathsf{ID}(p)$.

Corollary 5.1 *If $\mathcal{A}$ is an intersecting set system on $\mathbf{G}_n$, then $\mathcal{P}(\mathcal{A})$ is an intersecting family in the Bruhat ordering.*

Proof. This follows from Statement 3 of Claim 4.1 and Part (5) of Lemma 4.5. □

It is possible to construct a set system from a family of permutations. If $\mathcal{P}$ is a family of permutations, then $\mathsf{ID}(\mathsf{min}(\mathcal{P}))$ is a set system on $\mathbf{G}_n$ called the *generating set for $\mathcal{P}$*. The next three lemmas give properties of this set system.

Lemma 5.2 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be a maximal intersecting family. If $p \in \mathsf{min}(\mathcal{P})$, then $p = \pi(\mathsf{ID}(p))$.*

Proof. Let $p \in \mathsf{min}(\mathcal{P})$ and set $A = \mathsf{ID}(p)$. Then for any permutation $q \in \mathcal{P}$ we have that $\mathsf{ID}(q) \cap A \neq \emptyset$. Thus $\pi(A)$ intersects every $q \in \mathcal{P}$. Since $\mathcal{P}$ is maximal, $\pi(A) \in \mathcal{P}$. By definition of $\pi(A)$, we have $\pi(A) \preceq p$ and since p is minimal in $\mathcal{P}$, we have $p = \pi(A)$. □

Lemma 5.3 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be a maximal intersecting family and define $\mathcal{A} = \mathsf{ID}(\mathsf{min}(\mathcal{P}))$. Then $\mathcal{A}$ is an intersecting set system. Further, $p \in \mathcal{P}$ if and only if $A \subseteq \mathsf{ID}(p)$ for some $A \in \mathcal{A}$.*

Proof. The first statement is clear from Statement 3 of Claim 4.1.

It follows from the definition of $\mathcal{A}$ that if $p \in \mathcal{P}$ then $A \subseteq \mathsf{ID}(p)$ for some $A \in \mathcal{A}$. For any permutation p, if $A \subseteq \mathsf{ID}(p)$ for some $A \in \mathcal{A}$ then $\pi(A) \in \mathcal{P}$ and $\pi(A) \preceq p$. Since $\mathcal{P}$ is maximal, this implies that $p \in \mathcal{P}$. □

Lemma 5.4 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be an intersecting family and define $\mathcal{A} = \mathsf{ID}(\mathsf{min}(\mathcal{P}))$. Then $\mathcal{P} \subseteq \mathcal{P}(\mathcal{A})$. Moreover, if $\mathcal{P}$ is a maximal intersecting family, then $\mathcal{P} = \mathcal{P}(\mathcal{A})$.* □

The next lemma follows from from Part (1) of Claim 4.1. The lemma shows that ID is a weak order preserving map on the poset, so it preserves the min operation.

Lemma 5.5 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be a maximal intersecting family of permutations. Then $\mathsf{ID}(\mathsf{min}(\mathcal{P})) = \mathsf{min}(\mathsf{ID}(\mathcal{P}))$.* □

Let $\mathcal{A}$ be a system of subsets from an m-set. Then $\mathcal{A}$ has the *Sperner property* if, for all distinct $A, B \in \mathcal{A}$, it follows that $A \not\subseteq B$ and $B \not\subseteq A$ (i.e. $\mathcal{A}$ is an antichain). We call $\mathcal{A}$ a *maximal intersecting antichain*, if for any set $B \notin \mathcal{A}$, there is a set $A \in \mathcal{A}$ such that either $A \cap B = \emptyset$, $A \subseteq B$, or $B \subseteq A$. Lemma 5.5 implies that $\mathsf{ID}(\mathsf{min}(\mathcal{P}))$ has the Sperner property. Also, from Part (5) of Lemma 4.5, we can conclude the following.

Lemma 5.6 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be a maximal intersecting family and define $\mathcal{A} = \mathsf{ID}(\mathsf{min}(\mathcal{P}))$. Then $\mathcal{A}$ is a maximal intersecting antichain.*

Proof. We see from Corollary 5.3 that $\mathcal{A}$ is intersecting. By Lemma 5.5, we have $\mathcal{A} = \mathsf{min}(\mathsf{ID}(\mathcal{P}))$ which shows that $\mathcal{A}$ has the Sperner property.

Suppose that $\mathcal{A}$ is not a maximal intersecting antichain. Then there must be some $B \notin \mathcal{A}$ such that $\mathcal{A} \cup \{B\}$ is intersecting and has the Sperner property; in particular, there is no $A \in \mathcal{A}$ such that $A \subseteq B$. Let $p = \pi(B)$. Since no $A \in \mathcal{A}$ has the property that $A \subseteq \mathsf{ID}(p)$, we have that $p \notin \mathcal{P}$. If $\mathcal{P}$ is maximal, then $\overline{p}$ must be in $\mathcal{P}$. But then there exists an $A \in \mathcal{A}$ such that $A \subseteq \mathsf{ID}(\overline{p}) = \overline{B}$. But this implies that $A \cap B = \emptyset$ and $\mathcal{A} \cup \{B\}$ is not intersecting. This contradiction shows that $\mathcal{A}$ is a maximal intersecting antichain. □

At this point, to avoid unnecessary technicalities, we associate any set of integers with the set of generators having those integers as subscripts; e.g. $\{1, 3, 4\} \equiv \{g_1, g_3, g_4\}$.

There are many maximal intersecting set systems that are not maximal intersecting antichains. For example, consider the following system of r-subsets of $M = \{1, \ldots, m\}$ (with $m > 2r$) defined by

$$\mathcal{H}_{m,r,k} = \{H \in \binom{M}{r} \mid |H \cap \{1, \ldots, 2k-1\}| \geq k\}, \tag{2}$$

where $1 \leq k \leq r \leq (m-1)/2$. If $k = 1$, this is the canonical intersecting set. But the set system $\mathcal{H}_{m,r,k}$ is a maximally intersecting antichain if and only if $k = r$. If

$r > k$, consider the set $\{r, \ldots, m\}$, which intersects each set in $\mathcal{H}_{m,r,k}$. Since $r > k$, the set $\{r, \ldots, m\}$ is neither contained in, nor contains any of the sets from $\mathcal{H}_{m,r,k}$. The following sets are all examples of maximally intersecting antichains,

$$\begin{aligned}
\mathcal{A} =& \{A \in \binom{M}{r} \mid |A \cap \{1, 2, ..., 2r-1\}| \geq r\}, \\
\mathcal{B} =& \{12,\ 13,\ 14,\ 15, \cdots, 1m\ , 234 \cdots m\}, \text{ and} \\
\mathcal{C} =& \{123, 134, 145, 156, 126, 235, 245, 246, 346, 356\} \text{ on an 6-set.}
\end{aligned}$$

The sets below are not maximally intersecting antichains,

$$\begin{aligned}
\mathcal{D} =& \{A \in \binom{M}{r} \mid |A \cap \{1, 2, ..., 2k-1\}| \geq k\}, \quad \text{if } r > k, \text{ and} \\
\mathcal{E} =& \{A \in \binom{M}{r} \mid 1 \in A\}, \quad \text{if } r > 1.
\end{aligned}$$

The final theorem of this section is our characterization of the maximum intersecting families in the Bruhat lattice.

Theorem 5.7 *Let $\mathcal{P} \subseteq \mathscr{B}_n$ be an intersecting family. Then $\mathcal{P}$ is maximum if and only if $\mathcal{A} = \mathsf{ID}(\mathsf{min}(\mathcal{P}))$ is a maximal intersecting antichain.*

Proof. We have already seen that if $\mathcal{P}$ is maximum, then $\mathcal{A}$ is a maximal intersecting antichain. Assume that $\mathcal{A}$ is a maximal intersecting antichain and let $\mathcal{P}$ be the set of permutations generated by $\mathcal{A}$. We will show that for every permutation p, either $p \in \mathcal{P}$ or $\overline{p} \in \mathcal{P}$.

For a permutation $p \notin \mathcal{P}$ let $B = \mathsf{ID}(p)$. If B intersects every set in $\mathcal{A}$, then by maximality, $A \subseteq B$ or $B \subseteq A$ for some $A \in \mathcal{A}$. The former implies that $p \in \mathcal{P}(\mathcal{A}) = \mathcal{P}$, while the latter implies that $\mathsf{ID}(p) \subseteq \mathsf{min}(\mathsf{ID}(\mathcal{P}))$, a contradiction. If B does not intersect every set in $\mathcal{A}$, then $B \cap A = \emptyset$ for some $A \in \mathcal{A}$. This implies that $A \subseteq \overline{B}$ so $\overline{p} \in \mathcal{P}$. $\square$

From Corollary 5.1, the set $\mathcal{P}(\mathcal{H}_{m,r,k})$ (defined in Equation 2) is an intersecting set of permutations in the Bruhat ordering. There many different set systems that are isomorphic to $\mathcal{H}_{m,r,k}$, formed by permuting the underlying set. It is possible to define a set of intersecting permutations for each of these different, but isomorphic, set systems. For example, let $(m, r, k) = (6, 2, 2)$, then $\mathcal{H}_{6,2,2} = \{(1, 2), (1, 3), (2, 3)\}$, and $\mathcal{H} = \{(1, 3), (1, 5), (3, 5)\}$ is isomorphic to $\mathcal{H}_{6,2,2}$. Both set systems can be used to construct an intersecting set of permutations in the Bruhat order

$$\begin{aligned}
\mathcal{P}(\mathcal{H}_{6,2,2}) &= \mathsf{up}(\{321456, 214356, 143256\}), \\
\mathcal{P}(\mathcal{H}) &= \mathsf{up}(\{214356, 213465, 124365\}).
\end{aligned}$$

Both of these sets have size 360, but they are non-isomorphic families in $\mathscr{B}_n$. This can be seen since the ranks of 321456 and 143256 are 3, whereas the other permutations are of rank 2. One can see that separation in the sets in the set system is the key distinction here.

The final Corollary of this section follows from Theorem 5.7.

Corollary 5.8 *Let $\mathcal{H}$ be a set system isomorphic to $\mathcal{H}_{m,r,k}$, with $1 \leq k \leq r < (m-1)/2$. If $k = r$ then $\mathcal{P}(\mathcal{H}) = m!/2$ while, if $k < r$, then $\mathcal{P}(\mathcal{H}) < m!/2$.*

6 Erdős–Ko–Rado Theorem for levels in the Bruhat lattice

In this section we prove that an Erdős–Ko–Rado Theorem holds for the rth level of $\mathscr{B}_n$, provided that n is large relative to r. Recall that $\mathscr{B}_n(r)$ denotes the set of all permutations of rank r in the Bruhat lattice. If the size of any 1-intersecting family $\mathcal{P} \subseteq \mathscr{B}_n(r)$ is no larger than the largest set of permutations in $\mathscr{B}_n(r)$ in the upset of a single element in $\mathscr{B}_n(1)$, then we say that $\mathscr{B}_n(r)$ is 1-EKR, or simply that the level is EKR. It is clear that $\mathscr{B}_n(2)$ is EKR. In this section we give a proof that both $\mathscr{B}_n(3)$ and $\mathscr{B}_n(4)$ are EKR when n is large, and more generally that $\mathscr{B}_n(r)$ is EKR, provided that n is large relative to r.

Lemma 6.1 *Let $\mathcal{H} = \mathscr{B}_n(3)$ and $p \in \mathscr{B}_n(1)$. Then $|\mathsf{up}_{\mathcal{H}}(p)| = \binom{n-1}{2}$.*

Proof. Define the polynomial $[m] = (1 + x + \cdots + x^{m-1})$; then the generating function for the sizes of the ranks in $\mathscr{B}_n$ is

$$F(x) = [n]! = (1+x)(1+x+x^2)\cdots(1+x+x^2+\cdots+x^{n-1}).$$

From [9], the generating function for the sizes of the ranks in $\mathsf{up}(p)$, where p has rank one, is given by the generating function

$$\frac{F(x)}{1+x} = \frac{[n]!}{1+x} = (1+x+x^2)\cdots(1+x+x^2+\cdots+x^{n-1}).$$

Hence $|\mathsf{up}_{\mathcal{H}}(p)|$ is the coefficient of x^2 in $F(x)/(1+x)$. This coefficient is the same as that in $(1+x+x^2)^{n-2}$, which is $\binom{n-2}{2} + (n-2) = \binom{n-1}{2}$. □

We will use the following result of Holroyd, Spencer and Talbot [4] on EKR theorems for graphs. For any graph, there is a corresponding lattice on the independent sets of the graph ordered by inclusion. If the rth-level of this lattice has the 1-EKR property,

then the graph is said to have the *r-EKR property*. In other words, a graph has the r-EKR property if the size of any intersecting system of independent sets of size r is at most that of all independent sets of size r that contain some fixed vertex of the graph.

The graph P_n^k is the kth power of the path on n vertices $\{1, 2, \ldots, n\}$; two vertices a, b in P_n^k are adjacent if and only if $1 \leq |a-b| \leq k$. The next result is equivalent to the fact that graph P_n^k is r-EKR for $r \geq 1$. A *k-separated r-set* is a set in which any two elements (or, in the case of $\mathbf{G}_n$, any two subscripts) differ by at least k. An independent set in P_n^k is a k-separated set. A 1-separated set will simply be called a *separated set*.

Theorem 6.2 ([4]) *The largest intersecting system of separated r-sets from an m-set has size $\binom{m-r}{r-1}$ and consists of all the separated sets that contain some fixed point.* □

Theorem 6.3 *The set $\mathscr{B}_n(3)$ is EKR for $n \geq 7$.*

Proof. Let $\mathcal{P} \subseteq \mathscr{B}_n(3)$ be an intersecting family of maximum size. Let $\mathcal{P}_i = \{p \in \mathcal{P} \mid |\mathsf{ID}(p)| = i\}$ where $i \in \{1, 2, 3\}$. Clearly, $\mathcal{P} = \bigcup_{i=1}^{3} \mathcal{P}_i$.

If $p \in \mathcal{P}_1$, then every permutation in $\mathcal{P}$ must be above $\mathsf{ID}(p)$ for $p \in \mathcal{P}_1$. This means that every permutation in $\mathcal{P}$ has the single element in $\mathsf{ID}(p)$ in its inverse descents set and so $\mathcal{P}$ is a star. Thus we may assume that $\mathcal{P}_1 = \emptyset$. Now we will consider $\mathcal{P}_i$ for $i > 1$.

Any permutation p in $\mathcal{P}_3$ must have $\mathsf{ID}(p) = \mathsf{Inv}(p)$, since it is a rank 3 permutation with 3 inverse descents. This implies that for any $p \in \mathcal{P}_3$, the set $\mathsf{ID}(p)$ must be a separated set since $g_i, g_{i+1} \in \mathsf{ID}(p)$ forces $(i, i+1)$, $(i+1, i+2)$, and $(i, i+2)$ to be inversions. Further, every set $\mathsf{Inv}(p)$, for $p \in \mathcal{P}_3$, has multiplicity 1. Thus the set system $\{\mathsf{ID}(p) \mid p \in \mathcal{P}_3\}$ is an intersecting system of separated sets that has the same size as $\mathcal{P}_3$. By Theorem 6.2 (with $m = n-1$) we have $|\mathcal{P}_3| \leq \binom{n-4}{2}$.

Next consider $\mathcal{P}_2$ and define the set system $\mathcal{A}_2 = \{\mathsf{ID}(p) \mid p \in P_2\}$. Clearly, $\mathcal{A}_2$ is an intersecting 2-set system. If $|\mathcal{A}_2| \geq 4$, then $\mathcal{A}_2$ must contain four sets isomorphic to

$$\{12, 13, 14, 15\}.$$

Since the inverse descents for the permutations in $\mathcal{P}_2$ and $\mathcal{P}_3$ are sets of size no more than 3 that intersect all the sets in $\mathcal{A}_2$, this implies that all the permutations in both $\mathcal{P}_2$ and $\mathcal{P}_3$ must have a common element in their inverse descent. Thus if $|\mathcal{A}_2| \geq 4$, then $\mathcal{P}$ is a star. So we can assume that $|\mathcal{A}_2| \leq 3$. The multiplicity of any set in $\mathcal{A}_2$ is no more than 4, so $|P_2| \leq 12$.

So, if $\mathcal{P}$ is not a star, then

$$|\mathcal{P}| \leq 12 + \binom{n-4}{2} = \frac{1}{2}(n^2 - 9n + 44),$$

which is no more than $\binom{n-1}{2} = \frac{1}{2}(n^2 - 3n + 2)$, provided that $n \geq 7$. $\square$

Before we prove that $\mathscr{B}_n(4)$ is EKR, we need to determine the size of a star at the fourth level of the Bruhat lattice.

Lemma 6.4 *Let $\mathcal{H} = \mathscr{B}_n(4)$ and $p \in \mathscr{B}_n(1)$, then $|\mathsf{up}_{\mathcal{H}}(p)| = \frac{1}{6}(n^3 - 3n^2 + 2n - 6)$.*

Proof. As stated earlier, in [9] it is shown that for a $p \in \mathscr{B}_n(1)$, the size of $\mathsf{up}_{\mathcal{H}}(p)$ is given by the coefficient of x^3 in

$$\frac{[n]!}{x+1} = (1 + x + x^2)(1 + x + x^2 + x^3) \cdots (1 + x + x^2 + \cdots + x^{n-1}).$$

There are $\binom{n-3}{1}$ ways to select the term x^3 from one of the factors. There are $(n-2)(n-3)$ ways to select the term x from one factor and x^2 from another factor. Finally, there are $\binom{n-2}{3}$ to choose the term x from three different factors. Thus the coefficient of x^3 is

$$(n-3) + (n-2)(n-3) + \binom{n-2}{3} = \frac{1}{6}(n^3 - 3n^2 + 2n - 6).$$

$\square$

We will also need the following two technical lemmas.

Lemma 6.5 *The size of the maximum intersecting 3-set system with no separated sets is $3n - 8$. The size of the maximum intersecting 4-set system with no separated sets is $2n^2 - 16n + 35$.*

Proof. Let $\mathcal{A}$ be an intersecting 3-set system that contains no separated sets. If all the sets in $\mathcal{A}$ have a common element, say i. Then the maximum size of $\mathcal{A}$ is

$$\binom{n-1}{2} - \binom{n-4}{2}.$$

This follows since $\binom{n-1}{2}$ is the number of 3-sets that contain i, and we can always find at least $\binom{n-4}{2}$ separated 3-sets that contain i (this is the number of separated sets that contain either 2 or $n-1$).

If the sets in $\mathcal{A}$ do not all contain a common element, then, by the Hilton-Milner Theorem [5], the size of the largest non-canonical intersecting system of 3-sets has size

$$\binom{n-1}{2} - \binom{n-4}{2} + 1 = 3n - 8.$$

Therefore, in both cases, the set $\mathcal{A}$ is no larger than $3n-8$.

Similarly, let $\mathcal{A}$ be an intersecting 4-set system that contains no separated sets. If all the sets in $\mathcal{A}$ contain a common element, then $\mathcal{A}$ is no larger than

$$\binom{n-1}{3} - \binom{n-5}{3}.$$

This is seen by taking the total number of sets that contain the fixed element and subtracting the fewest number of separated sets that contain a common element (again this is the collection of all separated sets that contain either 2 or $n-1$).

By the Hilton-Milner Theorem the largest non-canonical intersecting set system has size

$$\binom{n-1}{3} - \binom{n-5}{3} + 1 = 2n^2 - 16n + 35.$$

□

Lemma 6.6 *Assume that $n > 31$. Let $\mathcal{F}$ be an intersecting family of 4-sets on an $(n-1)$-set with $|\mathcal{F}| > 9\binom{n-2}{2}$. Then $\mathcal{F}$ is a star and contains at least 4 sets that pairwise intersect in exactly the one common element.*

Proof. First, if $n \leq 31$ then the EKR theorem implies that no intersecting family of 4-sets is larger than $9\binom{n-2}{2}$.

By the Hilton-Milner theorem, if

$$|\mathcal{F}| > \binom{(n-1)-1}{3} - \binom{(n-1)-5}{3} + 1 = 2n^2 - 20n + 36,$$

then $\mathcal{F}$ is a star. We have that $|\mathcal{F}| > 9\binom{n-2}{2}$ which is larger than $2n^2 - 20n + 36$ for the given values of n. So we may assume that each set in $\mathcal{F}$ contains a common element.

Pick any set $A \in \mathcal{F}$, then at most $3\binom{n-2}{2}$ sets from $\mathcal{F}$ can intersect with A in more than one element. So if $|\mathcal{F}| > 9\binom{n-2}{2}$, then there are at least 4 sets in $\mathcal{F}$ that intersect only in the element that is common in all the sets of $\mathcal{F}$. □

Theorem 6.7 *The set $\mathscr{B}_n(4)$ is EKR, for $n \geq 42$.*

Proof. Let $\mathcal{P} \subseteq \mathscr{B}_n(4)$ be an intersecting family of maximum size. As in the proof of Theorem 6.3, let $\mathcal{P}_i = \{p \in \mathcal{P} \mid |\mathsf{ID}(p)| = i\}$ where $i \in \{1,2,3,4\}$. Clearly, $\mathcal{P} = \bigcup_{i=1}^{4} \mathcal{P}_i$.

If $\mathcal{P}_1 \neq \emptyset$ then the set system $\{\mathsf{ID}(p) \mid p \in \mathcal{P}_1\}$ is a star, so we may assume that $\mathcal{P}_1 = \emptyset$. Now we will consider $\mathcal{P}_i$ for $i > 1$.

Define the set system $\mathcal{A}_4 = \{\mathsf{ID}(p) \,|\, p \in \mathcal{P}_4\}$. Since all the permutations in $\mathcal{P}$ are rank 4, $|\mathcal{A}_4| = |\mathcal{P}_4|$. By Lemma 6.6, if $\mathcal{A}_4 > 9\binom{n-2}{2}$ and $n \geq 42$ then $\mathcal{A}_4$ is a star, and contains at least 4 sets that pairwise intersect at exactly one common element. Further, since all the other permutations in $\mathcal{P}$ have at most three inverse descents, to intersect all of the permutations in $\mathcal{P}_4$, the sets of inverse descents must intersect all four of the sets provided by Lemma 6.6. Thus the inverse descents must contain the same common element as the sets in $\mathcal{P}_4$. Hence, if $\mathcal{P}_4 > 9\binom{n-2}{2}$, then $\mathcal{P}$ must be a star. So we may assume that $\mathcal{P}_4 \leq 9\binom{n-2}{2}$.

Define the set system $\mathcal{A}_3 = \{\mathsf{ID}(p) \,|\, p \in P_3\}$. The sets in $\mathcal{A}_3$ can be either separated sets, or have two consecutive elements. The multiplicity of any separated set in $\mathcal{A}_3$ is at most 6 (since there must be exactly one more inversion) and by Lemma 6.2 there cannot be any more than $\binom{n-4}{2}$ such sets. The multiplicity of a set with two consecutive elements is one, since the two consecutive inversions result in three inversions. By Lemma 6.5 there are no more than $3n-8$ intersecting 3-sets with no separated sets. Thus the size of $\mathcal{P}_3$ is no more than

$$6\binom{n-4}{2} + (3n-8).$$

Next consider the set system $\mathcal{A}_2 = \{\mathsf{ID}(p) \,|\, p \in P_2\}$; this is an intersecting 2-set system. If $|\mathcal{A}_2| \geq 5$, then all the sets in $\mathcal{A}_2$ must contain a common element. Further every other permutation in $\mathcal{P}$ must have this element as an inverse descent, since the permutations in $\mathcal{P}$ have no more than four inverse descents. This implies that $\mathcal{P}$ must be a star. So we will assume that $|\mathcal{A}_2| \leq 4$. The multiplicity of any set in $\mathcal{A}_2$ is no more than 10, thus $|\mathcal{P}_2| \leq 40$.

So, if $\mathcal{P}$ is not a star, then

$$|\mathcal{P}| \leq 9\binom{n-2}{2} + 6\binom{n-4}{2} + (3n-8) + 40 = \frac{1}{2}(15n^2 - 93n + 237)$$

which is less than $\frac{1}{6}(n^3 - 3n^2 + 2n - 6)$ (which from Lemma 6.4 is the size of the upset of a single element intersected with level 4 in the weak Bruhat poset), provided that $n \geq 42$. □

We will next use a counting method, similar to the one used in Theorems 6.3 and 6.7, to show that the rth level of $\mathcal{B}_n$ is EKR, provided that n is sufficiently large. To start we will prove a lower bound on the size of a star in $\mathscr{B}_n(r)$.

Lemma 6.8 *Let $\mathcal{H} = \mathscr{B}_n(r)$ and $p \in \mathscr{B}_n(1)$, then $|\mathsf{up}_{\mathcal{H}}(p)| > \binom{n-2}{r-1}$.*

Proof. As we have seen, if $p \in \mathscr{B}_n(1)$, then the size of $\mathsf{up}_{\mathcal{H}}(p)$ is given by the coefficient of $x^{(r-1)}$ in

$$\frac{[n]!}{x+1} = (1+x+x^2)(1+x+x^2+x^3)\cdots(1+x+x^2+\cdots+x^{n-1}).$$

There are $\binom{n-2}{r-1}$ ways to select the term x from $r-1$ different factors. So the coefficient of $x^{(r-1)}$ is at least $\binom{n-2}{r-1}$. $\square$

Lemma 6.9 *Let $\mathcal{F}$ be an intersecting family of ℓ-sets on an $(n-1)$-set with $2 \leq \ell$. If $2 \leq r$ and*

$$|\mathcal{F}| > r(\ell-1)\binom{n-2}{\ell-2},$$

then $\mathcal{F}$ is a star and contains at least $r+1$ sets that pairwise intersect in exactly the one common element.

Proof. First note that since $2 \leq r, \ell$, it follows that

$$|\mathcal{F}| > \binom{(n-1)-1}{\ell-1} - \binom{(n-1)-(\ell+1)}{\ell-1} + 1,$$

and the Hilton-Milner Theorem implies that $\mathcal{F}$ is a star.

Pick any set $A \in \mathcal{F}$. At most $(\ell-1)\binom{n-2}{\ell-2}$ sets from $\mathcal{F}$ can intersect with A in more than one element. So if $|\mathcal{F}| > r(\ell-1)\binom{n-2}{\ell-2}$, then there are at least $r+1$ sets in $\mathcal{F}$ that intersect only in the element that is common in all the sets of $\mathcal{F}$. $\square$

Theorem 6.10 *The set $\mathscr{B}_n(r)$ is EKR, for n sufficiently large relative to r.*

Proof. We may assume that $r > 2$, since we have seen that $\mathscr{B}_n(2)$ is EKR. Let $\mathcal{P} \subseteq \mathscr{B}_n(r)$ be an intersecting family of maximum size. As in the previous theorems, let $\mathcal{P}_i = \{p \in \mathcal{P} \,|\, |\mathsf{ID}(p)| = i\}$ where $i \in \{1, 2, \ldots, r\}$. Clearly, $\mathcal{P} = \bigcup_{i=1}^{r} \mathcal{P}_i$. Further set $\mathcal{A}_i = \{\mathsf{ID}(p) \,|\, p \in P_i\}$.

As see have seen in the proofs of Theorem 6.3 and 6.7, if $\mathcal{P}_1 \neq \emptyset$ then then $\mathcal{P}$ is a star, so we may assume that $\mathcal{P}_1 = \emptyset$. Now we will consider $\mathcal{P}_\ell$ for $\ell \geq 2$.

By Lemma 6.9, if $|\mathcal{A}_\ell| > r(\ell-1)\binom{n-2}{\ell-2}$, then $\mathcal{A}_\ell$ is a star and contains at least $r+1$ sets that pairwise intersect at exactly the one common element. Further, each permutation in $\mathcal{P}$ has a set of no more than r inverse descents, and each set of inverse descents must intersect the $r+1$ sets provided by Lemma 6.9. So each set of inverse descents must contain the same common element that is in all the $\mathcal{A}_\ell$. Thus all the permutations in $\mathcal{P}$ must contain the same common inverse descent as the permutations in $\mathcal{P}_\ell$, and $\mathcal{P}$ is a star. Hence, we may assume that $\mathcal{A}_\ell \leq (r)(\ell-1)\binom{n-2}{\ell-2}$ for each $\ell \in \{2, \ldots, r\}$.

The multiplicity of any set from $\mathcal{A}_\ell$ in $\mathcal{P}_\ell$, is at most $\left(\!\binom{2\ell}{r-\ell}\!\right) = \binom{r+\ell-1}{r-\ell}$. Thus

$$\begin{aligned}|\mathcal{P}| &\leq \sum_{\ell=2}^{r} \binom{r+\ell-1}{r-\ell}(r)(\ell-1)\binom{n-2}{\ell-2} \\ &\leq (r-1)\binom{2r-1}{r}(r)(r-1)\binom{n-2}{r-2}.\end{aligned}$$

Provided that n is larger than

$$r(r-1)^3\binom{2r-1}{r} + r,$$

the size of $\mathcal{P}$ is less than $\binom{n-2}{r-1}$. □

The previous result uses a very rough counting method to bound the size of an intersecting family of permutations that is not a star. We believe that the lower bound on n given above is far larger than necessary, in fact we conjecture that all levels below the middle level in the weak Bruhat lattice have the EKR property.

Conjecture 6.11 *For all values of $1 \leq r \leq \frac{1}{2}\binom{n}{2}$ the set $\mathcal{B}_n(r)$ is EKR. Thus the size of the largest intersecting family of rank r permutations of $\{1, \ldots, n\}$ equals $|\mathsf{up}(p)|$, for any $p \in \mathcal{B}_n(1)$, which equals the coefficient of x^{r-1} in $F(x)/(1+x)$.*

7 t-intersecting families

In this section we conjecture that a version of the EKR theorem holds for some levels of the Bruhat lattice for t-intersecting permutations. The original version of the EKR theorem was for t-intersecting sets, and can be stated as follows.

Theorem 7.1 *Let k, t and n be positive integers with $t < k$. Assume that $\mathcal{A}$ is a t-intersecting system of k-subsets of $\{1, \ldots, n\}$. There exists a function $f(k,t)$ such that, if $n > f(k,t)$, then*

$$|\mathcal{A}| \leq \binom{n-t}{k-t}.$$

Moreover, equality holds if and only if $\mathcal{A}$ is a t-star at level k.

Similar to the EKR theorem for t-intersecting k-sets, we conjecture that the largest t-intersecting family of rank k permutations in the Bruhat lattice is the upset of a

permutation at level t. Unlike the case for sets, the sizes of the upsets of the permutations at level t are not all equal in the Bruhat lattice. Before stating the conjecture, we need to determine which permutation at level t we believe has the maximum number of rank r permutations in its upset.

Consider the permutation $\rho(t)$ that is formed by starting with the trivial permutation $\langle 1, 2, \ldots, n\rangle$ and reversing exactly t pairs. These reversals are done iteratively in t steps. At each step, the largest element in the permutation that has a smaller element directly before it (this is the largest element with a smaller element on its left), is switched with the element directly to the left of it. For example, for $n = 6$ we have

$$\begin{aligned}
\rho(0) &= 123456, & \rho(1) &= 123465, & \rho(2) &= 123645, & \rho(3) &= 126345,\\
\rho(4) &= 162345, & \rho(5) &= 612345, & \rho(6) &= 612354, & \rho(7) &= 612534,\\
\rho(8) &= 615234, & \rho(9) &= 651234, & \rho(10) &= 651243, & \rho(11) &= 651423,\\
\rho(12) &= 654123, & \rho(13) &= 654132, & \rho(14) &= 654312, & \rho(15) &= 654321.
\end{aligned}$$

If i and j are positive integers with i chosen as large as possible so that

$$t = (n-1) + (n-2) + \cdots + (n-i) + (n-(i+j+1)) \tag{3}$$

then

$$\rho(t) = \langle n, n-1, \ldots, n-i+1, 1, 2, \ldots, j, n-i, j+1, j+2, \ldots, n-i-1\rangle.$$

The set $\mathsf{Inv}(\rho(t))$ is the set of the final t transpositions in the lexicographic ordering (for example, if $n = 6$, then $\mathsf{Inv}(\rho(11)) = \{(2,4),(3,4),(1,5),\ldots,(4,5),(1,6),\ldots,(5,6)\}$) and $\rho(t)$ is at level t in the Bruhat lattice.

By Corollary 3.7 of [9], the generating function for the sizes of the levels in the down set of $\rho(t)$ is given by

$$[n][n-1]\cdots[n-i+1][n-i-j],$$

(where i and j are as defined in Equation 3). By Theorem 3.1 of [9] this implies that generating function of the levels in the upset of $\rho(t)$ is

$$F_{n,t}(x) \;:=\; [n-i]\cdots[n-i-j+1][n-i-j-2]\cdots[2][1].$$

For example, with $n = 4$ and $t = 4$ (see Figure 1), we have $\rho(4) = 4132$, $i = 1$, and $j = 1$. This yields the generating functions $[4][2] = 1 + 2x + 2x^2 + 2x^3 + 2x^3 + x^4$ for its downset and $F_{4,4}(x) = [3] = 1 + x + x^2$ for its upset.

We conjecture that $\rho(t)$ is a rank t permutation with the maximum number of rank r permutations in its upset.

Conjecture 7.2 *Let $1 \le t \le r \le \binom{n}{2}$. If n is sufficiently large relative to t and r, then the set $\mathcal{B}_n(r)$ is t-EKR. In particular, the size of the largest t-intersecting family of rank r permutations equals $|\mathsf{up}(p) \cup \mathcal{B}_n(r)|$, where $p = \rho(t)$; i.e. $f_{t,r}(n)$ equals the coefficient of x^{r-t} in $F_{n,t}(x)$.*

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